



STRIDER: Navigation via Instruction-Aligned Structural Decision Space Optimization

Abstract

The Zero-shot Vision-and-Language Navigation in Continuous Environments (VLN-CE) task requires agents to navigate previously unseen 3D environments using natural language instructions, without any scene-specific training. A critical challenge in this setting lies in ensuring agents' actions align with both spatial structure and task intent over long-horizon execution. Existing methods often fail to achieve robust navigation due to a lack of structured decision-making and insufficient integration of feedback from previous actions. To address these challenges, we propose STRIDER (Instruction-Aligned Structural Decision Space Optimization), a novel framework that systematically optimizes the agent's decision space by integrating spatial layout priors and dynamic task feedback. Our approach introduces two key innovations: 1) a Structured Waypoint Generator that constrains the action space through spatial structure, and 2) a Task-Alignment Regulator that adjusts behavior based on task progress, ensuring semantic alignment throughout navigation. Extensive experiments on the R2R-CE and RxR-CE benchmarks demonstrate that STRIDER significantly outperforms strong SOTA across key metrics; in particular, it improves Success Rate (SR) from 29% to 35%, a relative gain of 20.7%. Such results highlight the importance of spatially constrained decision-making and feedback-guided execution in improving navigation fidelity for zero-shot VLN-CE.

1 Introduction

VLN-CE challenges agents to follow natural language instructions to navigate previously unseen 3D environments without any scene-specific training or fine-tuning [42, 25, 5, 60, 35]. This task is a critical benchmark for embodied AI, requiring agents to generalize perception, reasoning, and action across diverse and dynamic scenes [3, 24, 40, 62, 47, 8]. In comparison to discrete VLN tasks [3, 51, 26], VLN-CE more closely reflects real-world deployment conditions, where agents must process RGB-D inputs and make continuous movement decisions [18, 61, 21]. As a result, zero-shot VLN-CE pushes the boundaries of language-grounded generalization in embodied navigation [38, 61, 35, 5].

A central challenge in VLN-CE lies not only in grounding instructions into perception, but also in ensuring that the agent's behavior remains aligned with the semantic intent of the instruction throughout the navigation process. In unfamiliar environments, agents may correctly understand the instruction yet still exhibit execution drift [47, 34, 42, 8, 18, 35], such as stopping near the target

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Figure 1: **Navigation behavior comparison between STRIDER and Open-Nav** [42]. Given the same instruction, Open-Nav demonstrates execution drift, such as prematurely turning away from a hallway, and accumulates deviations over time. In contrast, STRIDER generates trajectories that more accurately follow the intended path and reach the goal region.

room without entering or prematurely turning away from a hallway, as shown in Fig. 1. These failures highlight a significant gap between *what the agent understands* and *how it acts*. Our key insight is that effective zero-shot VLN-CE agents must go beyond strong perception and reasoning—they must operate within an *Instruction-Aligned Structural Decision Space*, a decision space that is explicitly structured by the environment and continuously regulated based on task progress.

However, existing approaches typically rely on learned waypoint predictors or sequence-to-sequence policies that map instructions and visual inputs directly to actions [42, 24, 18, 8]. While effective at modeling local navigability, these models tend to ignore structured representations that capture the global layout or semantic task progression [15, 16, 41, 26, 34]. Moreover, these methods often operate in an open-loop fashion, inferring each action independently without feedback on prior decisions [2, 9, 56, 38, 61]. This limitation hinders their ability to assess whether actions have brought the agent closer to the goal, leading to deviations from the instruction’s intent, especially in complex or ambiguous scenes. While instruction grounding has seen significant progress, insufficient attention has been paid to optimizing the agent’s decision space in a manner that aligns both with spatial structure and task instructions.

To address these challenges, we introduce **STRIDER**, a zero-shot VLN-CE framework built on the principle of *Instruction-Aligned Structural Decision Space Optimization*. Our approach is grounded in the observation that semantic misalignment often arises not from perceptual misunderstanding, but from the inability to maintain alignment with both the spatial structure of the environment and the task’s semantic progress over long-horizon execution [3, 23]. STRIDER adopts a modeling-first approach: instead of directly predicting actions from visual inputs and instructions, it focuses on optimizing the agent’s decision space by integrating spatial structure and task progress awareness [50]. By embedding spatial layout priors and continuous goal feedback into the agent’s decision-making process, STRIDER enables the agent to navigate within paths that are both spatially coherent and semantically aligned with the task. This dynamic adjustment of behavior reduces execution drift and improves instruction fidelity, enabling both spatial generalization and instruction-level adherence.

STRIDER achieves this optimization through two tightly integrated modules. The **Structured Waypoint Generator** creates a layout-constrained action space by extracting skeletons from depth-based navigable regions [10, 36, 45]. This module ensures that the agent’s movement decisions are limited to paths that are both spatially coherent and meaningful, grounded in the environment’s structure. The **Task-Alignment Regulator** continuously monitors task progress and adjusts the agent’s behavior accordingly, ensuring actions remain aligned with the instruction-defined goal. It recalibrates behavior whenever deviations are detected, while staying within the spatial constraints defined by the structured action space. Together, these modules optimize the agent’s decision space by structuring it with spatial constraints and regulating it according to task progress, ultimately enabling more efficient and semantically faithful navigation.

Our contributions are summarized as follows:

- We propose **STRIDER**, a zero-shot VLN-CE framework based on the principle of *Instruction-Aligned Decision Space Optimization*. STRIDER optimizes the agent’s decision space by integrating spatial structure and task feedback, enabling more coherent and instruction-faithful navigation behavior.
- STRIDER consists of two tightly coupled modules that jointly optimize the agent’s decision space: (1) A **Structured Waypoint Generator** that constructs a layout-constrained planning space from depth-based skeletons, embedding spatial priors into the action space; (2) A **Task-Alignment Regulator** that monitors semantic progress across steps and adjusts behavior accordingly, ensuring actions remain aligned with instruction goals.
- We evaluate STRIDER on two standard zero-shot VLN-CE benchmarks, R2R-CE and RxR-CE, where it consistently outperforms strong baselines on core navigation metrics, demonstrating the benefit of structuring and regulating the agent’s decision space.

2 Related Work

2.1 Vision-and-Language Navigation

Significant progress has been made in Vision-and-Language Navigation (VLN), especially in continuous settings [42, 24, 25, 18, 39, 3]. VLN approaches can be broadly categorized into supervised learning with environment-specific training and zero-shot generalization. Supervised methods, such as imitation learning and reinforcement learning [53], rely on human-annotated trajectories to train navigation policies [2, 39, 13, 58, 59, 19], often enhancing performance through visual-language alignment, attention modules, auxiliary tasks [39, 50, 62, 1], and explicit or implicit 3D scene understanding and reconstruction [55, 31, 29, 30, 14, 32]. On the other hand, zero-shot methods aim to generalize to unseen scenes without task-specific fine-tuning. These approaches typically leverage instruction tuning, pretrained Vision-Language Models (VLMs) or Large Language Models (LLMs) [42, 37, 46, 54, 48]. While zero-shot methods show promise, they primarily focus on local observations and immediate actions, lacking global spatial awareness and long-term planning capabilities. Few methods integrate feedback mechanisms to monitor task progress or correct deviations, limiting their adaptability in complex scenarios.

2.2 Decision Space Optimization

Decision Space Optimization originates from robotics and autonomous systems, where it involves structuring an agent’s decision-making within a "decision space" that integrates spatial constraints, task goals, and feedback from the environment [27, 52, 22, 33]. This structured decision-making allows agents to balance short-term actions with long-term objectives, improving task execution. In robotics, decision space optimization ensures that actions are spatially coherent and consistent with global goals, aiding navigation and manipulation in dynamic environments [49, 43, 11]. In the field of VLN, early methods primarily map instructions to actions based on local observations but struggle with long-term planning and generalization to unseen environments [3, 13, 39]. More recent methods have incorporated memory, attention mechanisms, and reinforcement learning to improve decision-making [53, 17, 41]. These approaches also use visual-language alignment for action prediction and introduce curriculum learning for enhanced performance [62, 2, 50]. However, challenges remain in maintaining global spatial awareness and ensuring long-term task alignment. STRIDER addresses these issues through Instruction-Aligned Structural Decision Space Optimization. It optimizes decision-making by ensuring actions are consistent with both spatial constraints and task goals, enabling better navigation performance over long horizons.

3 Method

In this section, we explore how VLN-CE can be enhanced by structuring the decision space through spatial constraints and integrating task feedback. We begin with an overview of the VLN-CE task and introduce our proposed STRIDER framework (Sec. 3.1). We then present two core components of our approach: (1) the use of local perception to construct spatially organized representations that capture region connectivity (Sec. 3.2), and (2) the integration of task feedback to monitor subgoal progress and regulate action selection during execution (Sec. 3.3).

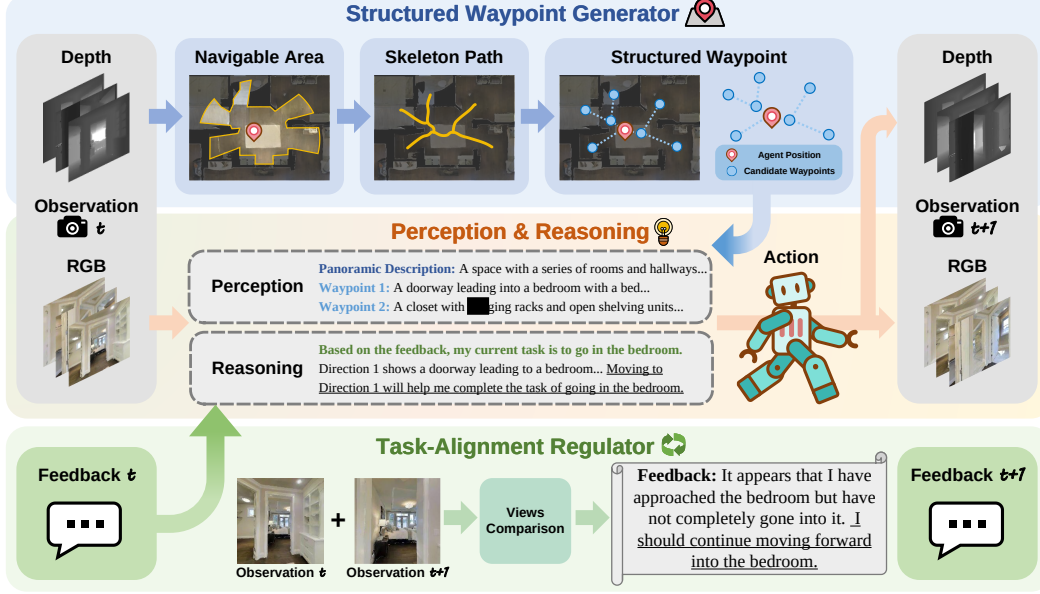


Figure 2: **Overview of the STRIDER pipeline.** The Structured Waypoint Generator constructs a layout-constrained waypoint space by extracting skeleton paths from navigable depth observations. The agent performs perception and reasoning over visual descriptions and feedback to identify suitable actions in context. To maintain semantic alignment over time, the Task-Alignment Regulator compares current and previous observations and generates feedback that guides the next action.

3.1 Overview

Task Definition. In the task of zero-shot VLN-CE, the agent begins at a specified starting location and must reach a target destination by interpreting both verbal guidance and visual observations. At each timestep, it receives a panoramic 360° view composed of 12 RGB-D images captured at fixed intervals (0°, 30°, ..., 330°). From these observations, the agent predicts a set of waypoints, candidate navigable locations, and selects one to move toward. The episode continues until the instruction is fulfilled or the agent reaches the goal.

STRIDER. Our proposed STRIDER framework follows a zero-shot VLN-CE pipeline, as illustrated in Fig. 2. At each timestep t , the agent receives an RGB-D observation $\mathbf{O}_t = (\mathbf{I}_t, \mathbf{D}_t)$ and a language instruction L . The Structured Waypoint Generator processes the depth $\mathbf{D}_t$ to extract navigable regions and generates a set of structured waypoints $\mathcal{W}_t$ organized by spatial connectivity. Subsequently, a pretrained Vision-Language Model (VLM) describes the RGB input $\mathbf{I}_t$, attending to visual content in the directions of candidate waypoints $\mathcal{W}_t$ to form a decision space $\mathcal{A}_t$. A Large Language Model (LLM) then reasons over the instruction L , the current decision space $\mathcal{A}_t$, and the task feedback f_t generated from the previous step to select a waypoint $w_t^* \in \mathcal{W}_t$ for movement.

After executing the action toward w_t^* , the agent receives the next observation $\mathbf{O}_{t+1}$. The Task-Alignment Regulator compares $\mathbf{O}_t$ and $\mathbf{O}_{t+1}$ using the VLM to detect progress toward the instruction goal and generates updated feedback f_{t+1} . This feedback is leveraged in the next decision step, completing a closed-loop control cycle grounded in structured perception and task alignment.

3.2 Structured Waypoint Generator

To optimize the agent’s decision space in continuous environments, we generate a layout-constrained set of candidate actions that explicitly reflect the spatial structure of the scene. Rather than relying on local navigability or unconstrained policy outputs, we propose a **Structured Waypoint Generator** that transforms depth input into a compact topological graph of navigable options. This process consists of three stages: (1) navigable region extraction, (2) topological skeleton abstraction, and (3) structured waypoint selection.

Navigable Region Extraction. At time step t , the agent receives a panoramic RGB-D observation $\mathbf{O}_t = (\mathbf{I}_t, \mathbf{D}_t)$, where $\mathbf{D}_t$ denotes multi-view depth input. Then we reconstruct a local point cloud $\mathcal{P}_t \subset \mathbb{R}^3$, where each point $\mathbf{p}_i = (x_i, y_i, z_i)$ represents a 3D coordinate in the agent-centric reference frame. To isolate feasible movement areas, we filter for ground-level points and project them into a 2D top-down plane:

$$\Omega_t = \Pi(\{\mathbf{p}_i \in \mathcal{P}_t \mid \mathbf{p}_i(z) < \delta_h, \|\mathbf{p}_i(x, y)\| < r\}), \quad (1)$$

where δ_h is the height threshold, r is the local planning radius, and $\Pi(\cdot)$ denotes orthographic projection onto the horizontal plane. Ω_t defines the local traversable area around the agent.

Topological Skeleton Extraction. To introduce structural priors into the local decision space, we abstract the raw traversable region Ω_t into a topological skeleton $\mathcal{S}_t$ using morphological thinning [28]:

$$\mathcal{S}_t = \text{Skeletonize}(\Omega_t), \quad (2)$$

where $\mathcal{S}_t \subset \Omega_t$ approximates the center axis of free space. Rather than interpreting navigable space as a dense, unstructured area, the skeleton captures its underlying topology by tracing the central lines of movement through open regions. This is analogous to how humans often form mental maps of environments based on key corridors, intersections, and doorways, rather than memorizing complete spatial coverage. By reasoning over this abstract structure, the agent can better plan paths that respect the environment’s layout and reduce unnecessary action noise.

Structured Waypoint Selection. We model the skeleton $\mathcal{S}_t$ as an undirected graph $\mathcal{G}_t = (\mathcal{V}_t, \mathcal{E}_t)$, where nodes $\mathcal{V}_t$ correspond to skeleton pixels and edges $\mathcal{E}_t$ reflect 8-connected adjacency (including horizontal, vertical, and diagonal neighbors in the 2D grid). To reduce redundancy and focus on directionally meaningful locations, we select a sparse set of endpoints from the graph as candidate waypoints. Specifically, we retain only nodes with degree 1:

$$\mathcal{W}_t = \{v_i \in \mathcal{V}_t \mid \deg(v_i) = 1\}, \quad (3)$$

which correspond to the outermost reachable points on the local skeleton, as shown in Fig. 3. These endpoints naturally capture the agent’s forward navigability and latent path divergence. Although we do not explicitly select junctions, their future branches are represented by the endpoints along each subpath.

Each selected waypoint $w_i \in \mathcal{W}_t$ is projected back into 3D. We input the RGB views $\mathbf{I}_t$ into a VLM to extract semantic information, focusing on visual content in the direction of each waypoint. For each w_i , the VLM outputs a textual description $\mathcal{D}_i$ of the corresponding direction. The final decision space is defined as a set of paired spatial-semantic candidates:

$$\mathcal{A}_t = \{(w_i, \mathcal{D}_i) \mid w_i \in \mathcal{W}_t, \mathcal{D}_i = \text{VLM}(\mathbf{I}_t, w_i)\}, \quad (4)$$

which couples structural layout with perceptual grounding. This layout-constrained action space forms part of the input for the LLM.

3.3 Task-Aligned Feedback Regulation

To ensure instruction-aligned behavior over long-horizon trajectories, we introduce a feedback regulation mechanism that dynamically adjusts the agent’s decision space based on recent observations and subtask progression. This module plays a central role in our principle of *Instruction-Aligned Decision Space Optimization*, allowing the agent to refine its actions not only based on spatial layout but also on semantic alignment with the instruction over time.

Visual Feedback Generation. Through LLM reasoning, the agent selects a waypoint w_t^* from the decision space $\mathcal{A}_t$, resulting in an action a_t . After executing a_t , it receives the next observation

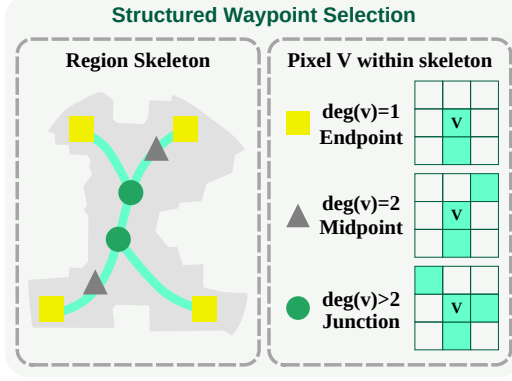


Figure 3: **Structured waypoint selection based on skeleton.** We categorize skeleton nodes by their degree and select only degree-1 endpoints as candidate waypoints.

$\mathbf{O}_{t+1} = (\mathbf{I}_{t+1}, \mathbf{D}_{t+1})$. To assess whether the agent has made progress toward the current subtask $\mathcal{T}_t$ (derived from instruction L), we input the observation pair $(\mathbf{O}_t, \mathbf{O}_{t+1})$ and the subtask $\mathcal{T}_t$ into the VLM to generate a feedback signal:

$$f_{t+1} = \text{VLM}(\mathbf{O}_t, \mathbf{O}_{t+1}, \mathcal{T}_t). \quad (5)$$

where f_{t+1} is a textual reflection describing the change in scene with respect to $\mathcal{T}_t$ —e.g., “partially entered the bedroom” or “moved away from the target.” The feedback offers fine-grained progress estimation at each step, allowing the agent to detect incremental advances or errors.

Feedback-Guided Action Selection. To determine the next action, the agent reasons over the updated structured decision space $\mathcal{A}_{t+1}$ (similarly defined in Eq. (4)), the full instruction L , and the generated feedback f_{t+1} . The LLM outputs the next action via:

$$w_{t+1}^* = \text{LLM}(\mathcal{A}_{t+1}, f_{t+1}, L), \quad (6)$$

$$a_{t+1} = \text{Action}(w_{t+1}^*). \quad (7)$$

This loop forms a closed decision-feedback cycle, in which action selection is continuously guided by semantic progress monitoring. As the agent moves through the environment, each observation is not only encoded visually but interpreted in light of task intent, enabling corrective adjustments and reducing cumulative drift. In this way, the decision space is adaptively regulated by execution context, maintaining semantic coherence with the instruction over time. By iterating this closed-loop process for T steps, the agent produces an action sequence $\{a_1, a_2, \dots, a_T\}$ that successfully navigates toward the instruction-aligned target.

4 Experiments

4.1 Experimental Setup

R2R-CE Dataset. We conduct experiments on the R2R-CE dataset, which extends the Room-to-Room (R2R) benchmark for visual language navigation (VLN) [3, 25]. This dataset consists of natural language instructions paired with navigation trajectories in realistic 3D indoor environments, derived from the Matterport3D dataset [4]. We follow the settings of OpenNav [42], conducting tests on 100 randomly selected episodes from the dataset. In these experiments, we leverage both VLM and LLM to perform zero-shot navigation. Our goal is to fully leverage the generalization and reasoning capabilities of these pretrained models, enabling them to adapt to the current task without any additional training.

RxR-CE Dataset. We also use the RxR-CE dataset, which extends the Room-Across-Room (RxR) benchmark with similar challenging conditions [26, 25]. RxR features longer and more diverse instructions across multiple languages and emphasizes global navigation capabilities. The CE variant (Continuously Evolving) simulates viewpoint changes and environmental variations, making it suitable for evaluating the robustness and generalization of navigation agents under distribution shifts.

Evaluation metrics. We evaluate navigation performance using standard metrics. Navigation Error (NE) measures the shortest-path distance between the agent’s final position and the goal. Success Rate (SR) is the percentage of episodes where the agent stops within 3 meters of the goal. Success weighted by Path Length (SPL) balances success and path efficiency [3]. Normalized Dynamic Time Warping (NDTW) reflects the similarity between the predicted and reference trajectories [20]. Oracle Success Rate (OSR) indicates the best possible success assuming the agent stops optimally along its path [25]. Trajectory Length (TL) records the average length of agent trajectories. Soft-DTW (SDTW) is a relaxed version of DTW that tolerates slight deviations in trajectory matching [12].

Implementation details. All experiments are conducted in simulated VLN-CE environments. At each step, the agent receives an RGB-D observation, where the RGB input is resized to $244 \times 244 \times 3$ and the depth map to 256×256 . Structured waypoints are generated by extracting skeletons from depth without relying on any pretrained waypoint predictor. For perception and feedback generation, we use Qwen-VL-Max as the Vision-Language Model (VLM). The action selection process is guided by GPT-4o, which reasons over the instruction, structured perception, and feedback to choose the next waypoint. Our VLM and LLM are accessed via API rather than deployed locally; for local deployment using open-source models, please refer to Open-Nav [42].

Table 1: **Performance comparison on the R2R-CE dataset.** The dash "-" indicates that the corresponding metric was not reported in the original work. **Bold** indicates the best result. We report the relative percentage change of our method compared to the previous SOTA in parentheses. **Red** denotes improvement, while **Green** indicates degradation.

Method	NE↓	NDTW↑	OSR↑	SR↑	SPL↑
Supervised Learning					
CMA [25]	6.92	50.77	45	37	32.17
RecBERT [57]	5.8	54.81	57	48	43.22
BEVBert [1]	5.13	61.40	64	60	53.41
ETPNav [2]	5.15	61.15	58	52	52.18
HNR [55]	4.42	-	67	61	51
Zero-Shot					
Random [42]	8.63	34.08	12	2	1.50
LXMERT [25]	10.48	18.73	22	2	1.87
DiscussNav-GPT4 [38]	7.77	42.87	15	11	10.51
Open-Nav-Llama3.1 [42]	7.25	44.99	23	16	12.90
Open-Nav-GPT4 [42]	6.70	45.79	23	19	16.10
SmartWay [46]	7.01	-	51	29	22.46
Ours	6.91(3.1%)	51.8(13.2%)	39(23.5%)	35(20.6%)	30.30(34.9%)

Table 2: **Performance comparison on the RxR-CE dataset.** The dash "-" indicates that the corresponding metric was not reported in the original work. **Bold** indicates the best result. We report the relative percentage change of our method compared to the previous SOTA in parentheses. **Red** denotes improvement, while **Green** indicates degradation.

Method	NE↓	NDTW↑	SDTW↑	SR↑	SPL↑
Supervised Learning					
LAW [44]	11.04	37.0	8.0	10.0	9.0
VLN-CBERT [19]	8.98	46.7	-	27.1	23.7
GridMM [56]	8.42	48.2	33.7	36.3	30.1
ETPNav [2]	5.64	61.9	45.3	54.8	44.9
WS-MGMap [6]	9.83	-	-	15.0	12.1
HNR [55]	5.51	63.56	47.24	56.39	46.73
Zero-Shot					
A ² Nav [7]	-	-	-	16.8	6.3
CA-Nav [5]	10.37	13.5	5.0	19.0	6.0
Ours	11.19(7.9%)	30.1(122.9%)	8.9(78.0%)	21.2(11.5%)	9.6(52.3%)

4.2 Main Results on Zero-Shot VLN-CE

We evaluate STRIDER on two standard zero-shot VLN-CE benchmarks: R2R-CE (Tab. 1) and RxR-CE (Tab. 2). Across both datasets, STRIDER consistently outperforms prior zero-shot methods on key metrics such as SPL, SR, and NDTW, indicating more reliable goal completion and higher trajectory fidelity. While supervised methods benefit from task-specific training, STRIDER remains competitive despite operating without fine-tuning or environment-specific adaptation.

On R2R-CE, STRIDER achieves substantial improvements in SPL and NDTW over all prior zero-shot models, driven by two key design factors. The Structured Waypoint Generator constrains the agent’s behavior to layout-consistent paths, reducing detours and spatial drift, while the Task-Alignment Regulator provides real-time feedback to maintain semantic alignment and correct deviations. This combination enables STRIDER to balance spatial feasibility and instruction adherence, leading to gains across both path-quality and goal-completion metrics. On the more diverse RxR-CE benchmark, STRIDER continues to outperform other zero-shot methods, though with narrower margins—likely due to RxR’s higher linguistic and geographic variability. Even in such settings, STRIDER’s structured decision space offers a strong prior that compensates for ambiguity, while the feedback

Table 3: **Effect of Structured Waypoint Generator (SWG) under different node degree configurations.** **Bold** indicates the best result. We report the relative percentage change compared to the baseline in parentheses. **Red** denotes improvement, while **Green** indicates degradation. The gray-shaded row denotes the primary experimental configuration used in our main results.

SWG	Node Degree	NE↓	NDTW↑	OSR↑	SR↑	SPL↑
✗	✗	7.19	48.78	29	24	21.07
✓	1	6.91(3.9%)	51.87(6.3%)	39(34.5%)	35(45.8%)	30.30(43.8%)
✓	> 2	7.34(2.1%)	49.88(2.3%)	33(13.8%)	28(16.7%)	25.02(18.7%)
✓	≠ 2	6.83(5.0%)	51.12(4.8%)	38(31.0%)	33(37.5%)	29.21(38.6%)

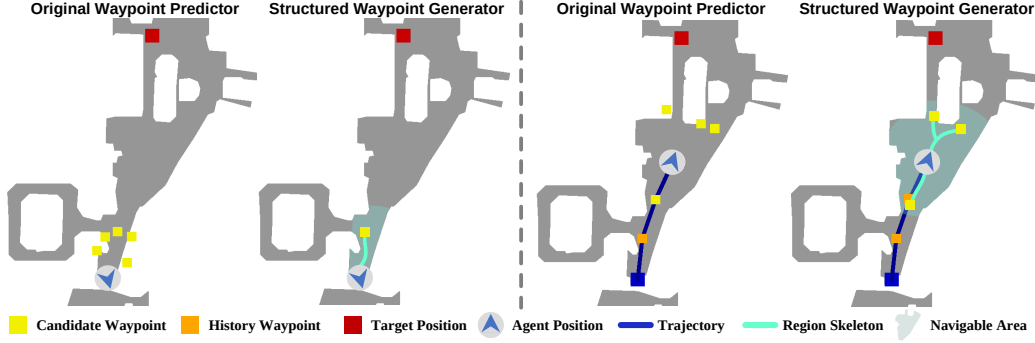


Figure 4: **Comparison between original waypoint predictor and Structured Waypoint Generator.** Our Structured Waypoint Generator extracts layout-consistent waypoints that align with the environment’s topology, resulting in trajectories that are more goal-directed and spatially coherent.

loop supports adaptive planning. Together, these elements demonstrate the robustness and generality of decision space optimization under zero-shot conditions.

It is worth mentioning that STRIDER does not achieve the lowest Navigation Error (NE), which is expected given its emphasis on instruction alignment and structural feasibility over exact endpoint proximity. The agent may stop at semantically appropriate locations slightly offset from the goal, reflecting a preference for coherent, interpretable paths over shortcut-based precision—an acceptable tradeoff in instruction-guided navigation tasks.

4.3 Ablation Study

We conduct ablation experiments to analyze the impact of STRIDER’s two core modules: the Structured Waypoint Generator and the Task-Alignment Regulator. For each module, we present both quantitative comparisons and qualitative visualizations to highlight their effects on navigation performance.

4.3.1 Effect of Structured Waypoint Generator

We evaluate different configurations of the Structured Waypoint Generator to understand how layout-aware guidance affects agent behavior. As shown in Tab. 3, we consider two key design choices: (1) whether to use the Structured Waypoint Generator, and (2) how to guide waypoint selection based on skeleton connectivity—e.g., using only endpoint nodes ($\deg(v_i)=1$), higher-degree junctions ($\deg(v_i)>2$) or both. We use the original waypoint predictor trained on the R2R dataset as the baseline for STRIDER without the Structured Waypoint Generator. We present the visualization comparison between the original waypoint predictor and the Structured Waypoint Generator in Fig. 4.

Using degree-1 nodes (endpoints) yields the best overall performance across all metrics, as these points typically lie at corridor tips or face the goal direction, providing clear guidance with minimal ambiguity. In contrast, using only high-degree nodes (>2), such as junctions, leads to degraded performance with higher NE (7.34) and lower SR/SPL, likely due to increased behavioral uncertainty.

Table 4: **Effect of the Task-Alignment Regulator (TAR) on navigation performance.** Bold indicates the best result. We report the relative percentage change compared to the baseline in parentheses. Red denotes improvement, while Green indicates degradation. The gray-shaded row denotes the primary experimental configuration used in our main results.

TAR	NE↓	NDTW↑	OSR↑	SR↑	SPL↑
✗	6.77	51.06	42	29	26.11
✓	6.91(2.1%)	51.87(1.6%)	39(7.1%)	35(20.7%)	30.30(16.0%)



Figure 5: **Comparison of agent behavior under no-feedback and feedback-driven execution strategies.** Without feedback, the agent prematurely infers task completion, resulting in an incorrect action (Action 2). With feedback, the agent leverages the intermediate state to refine its understanding, yielding a more semantically consistent action (Action 1) aligned with the instruction.

Combining non-2-degree nodes (i.e., $\neq 2$) provides a good compromise, incorporating both endpoints and informative junctions to achieve strong performance (NE: 6.83, SPL: 29.21) at the cost of a slightly longer trajectory. Based on these results, we adopt the degree-1 configuration as the default in our main experiments.

4.3.2 Effect of Task-Alignment Regulator

As shown in Tab. 4, enabling the Task-Alignment Regulator (TAR) leads to consistent improvements across SPL, SR, and NDTW, indicating that feedback-driven behavior adjustment helps the agent maintain semantic alignment and recover from drift during long-horizon navigation. While NE slightly increases (6.77→6.91), this reflects a more conservative execution pattern, where the agent prioritizes instruction fidelity and avoids premature termination, ultimately leading to higher task success and trajectory consistency. We illustrate the impact of the Task-Alignment Regulator on navigation behavior in Fig. 5. Overall, TAR enhances instruction fidelity without compromising spatial plausibility and is adopted in our main model configuration.

4.3.3 Effect of Model-Agnostic Design

STRIDER is not tied to any single model and can operate effectively across components of similar capabilities. To verify this, we conduct experiments using various VLMs of different sizes and providers. As shown in Tab. 5, GPT-4o achieves the best performance in terms of NE (6.75) and SR (36). However, STRIDER also shows competitive results using similar models, such as Qwen-VL-Max and Qwen2.5-VL-72B, which perform strongly in NDTW, OSR and SR, while still maintaining an overall solid performance. Additionally, STRIDER continues to deliver good results with smaller models like Qwen2.5-VL-32B and Qwen2.5-VL-7B. The consistent performance across different

Table 5: **Ablation on different VLMs.** We test our method using various VLMs of different sizes and capabilities. **Bold** indicates the best result. The gray-shaded row denotes the primary experimental configuration used in our main results.

VLM	TL	NE↓	NDTW↑	OSR↑	SR↑	SPL↑
Qwen-VL-Max	8.13	6.91	51.87	39	35	30.30
Qwen2.5-VL-72B	8.30	6.78	51.99	39	34	29.07
Qwen2.5-VL-32B	8.56	7.12	48.02	33	28	24.20
Qwen2.5-VL-7B	8.92	7.46	46.35	29	24	21.12
GPT-4o	8.01	6.75	50.12	39	36	31.37
Gemini-2.5-Pro	8.34	6.92	51.35	37	34	29.85
Gemini-2.5-Flash	7.68	7.08	49.87	34	29	25.30
Claude-3.5	7.81	6.86	52.10	36	33	29.40
Claude-4	8.22	7.14	45.25	31	29	26.10

Table 6: **Applying SWG to BEVBert.** We compare the vanilla BEVBert with the one applying our SWG. **Bold** indicates the best result. We report the relative percentage change compared to the baseline in parentheses. **Red** denotes improvement.

Method	NE↓	OSR↑	SR↑	SPL↑
BEVBert	4.57	67	59	50
BEVBert w/ SWG	4.37(4.3%)	70(4.4%)	61(3.3%)	53(6.0%)

models suggests that STRIDER’s design amplifies the strengths of the underlying model without relying on a specific model.

For our primary experiments, we selected Qwen-VL-Max. This choice highlights STRIDER’s model-agnostic design, which enables our method to work with any similar model and ensures that its results are driven by the strength of our approach, rather than reliance on a specific foundation model.

4.3.4 Applying SWG to Fine-Tuned Models

We further assess the effectiveness of Structured Waypoint Guidance (SWG) in fine-tuned models. Specifically, we incorporate our SWG module into the BEVBert model by replacing its original waypoint prediction mechanism. This modification leads to improvements across all key evaluation metrics, as shown in Table 6.

Even in fine-tuned settings, SWG’s ability to integrate environmental structure as a strong prior helps compensate for uncertainties in the navigation task. This further reinforces the versatility and effectiveness of SWG in different contexts, demonstrating that structured priors can be seamlessly integrated into existing models, thereby improving their robustness and reliability across a range of navigation tasks.

5 Conclusion

We presented STRIDER, a zero-shot VLN-CE framework that optimizes agent behavior through Instruction-Aligned Structural Decision Space Optimization. By combining a Structured Waypoint Generator with a Task-Alignment Regulator, STRIDER enables agents to navigate in complex, unseen environments in a manner that is both structurally coherent and semantically faithful to natural language instructions. Extensive experiments on VLN-CE benchmarks demonstrate that our approach outperforms strong zero-shot baselines across multiple metrics, with ablations confirming the complementary contributions of structural planning and feedback-driven regulation. This work highlights the importance of structuring and modulating the decision space for long-horizon instruction following and opens up future directions in integrating richer spatial priors, adaptive subgoal modeling, and instruction-aware exploration strategies.

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